# PASSWORD CRACKING (ON OPEN-SOURCE SYSTEM)

**Objective:** Recover user's passwords from an intentionally vulnerable open-source system using john the ripper and hashcat.

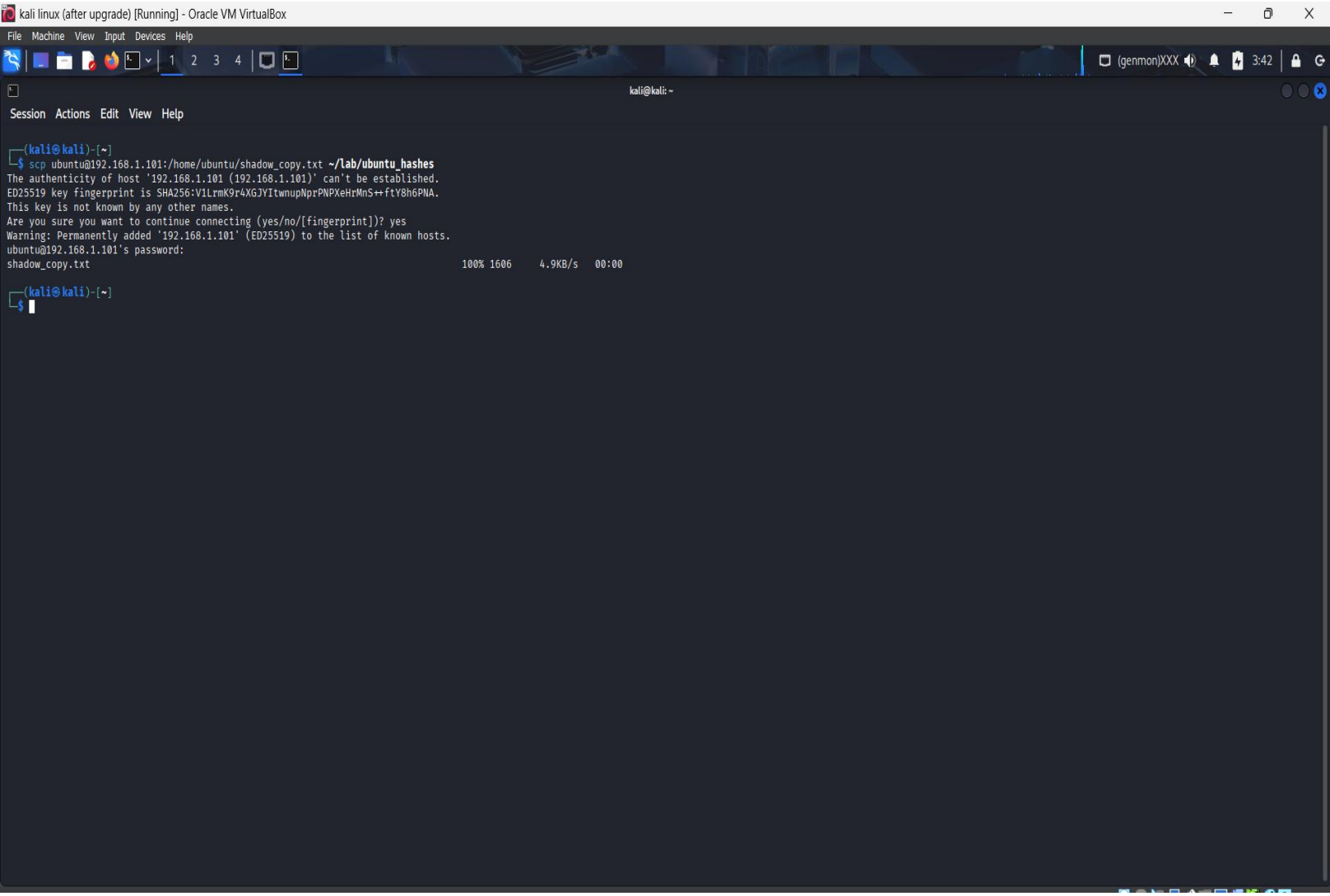
**Target:** Ubuntu (linux)

**Attack type:** Offline cracking (dictionary attack)

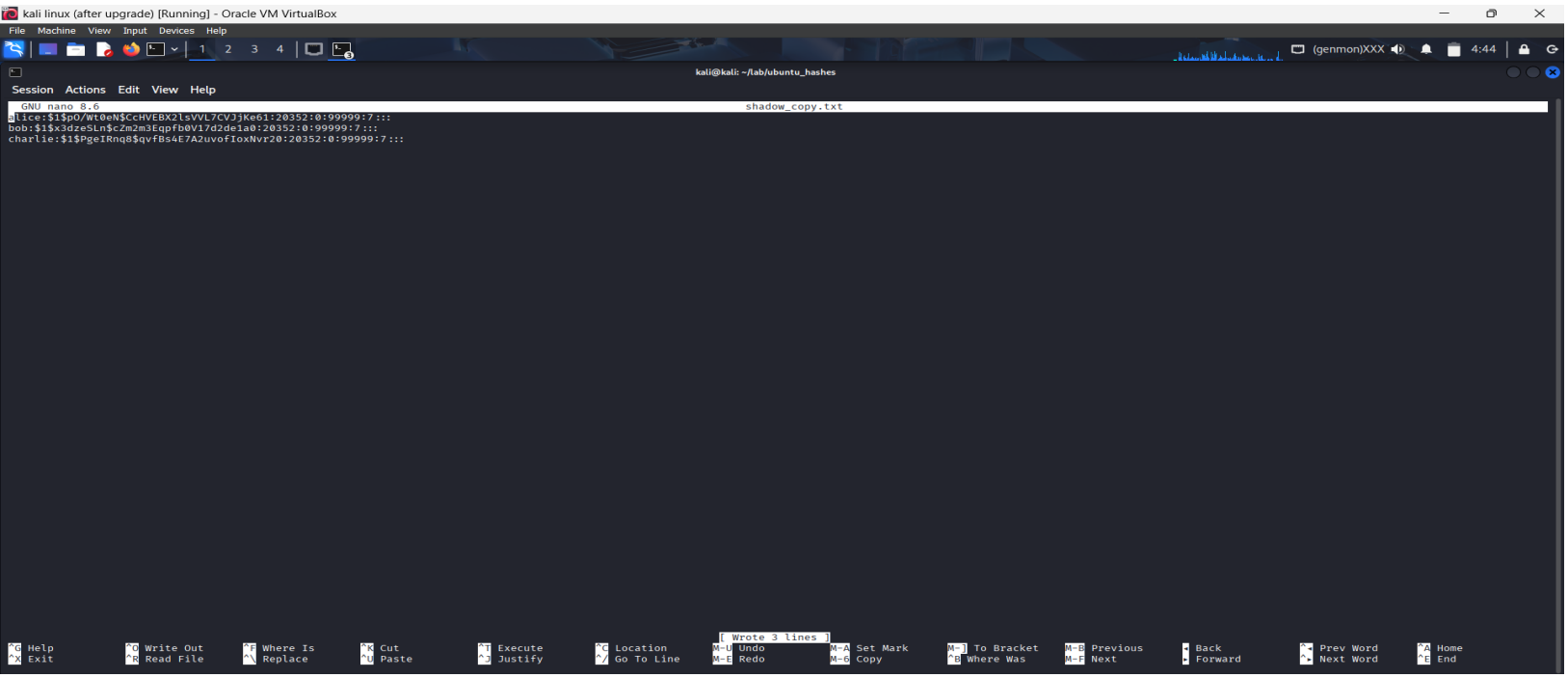
**Environmental setup:**

1. Host system: Kali Linux (VM) on virtual box
2. Target VM: Ubuntu
3. Network: an internal LAN network so attack VM cannot reach the internet or external network
4. Tools preinstalled on kali:
   John (john the ripper): it is fast, flexible and good for many hashes types.
   Hashcat: it is GPU-accelerated and support many hash modes and large combinatorics.
   rockyou.txt(wordlist)

I connected to the target machine remotely and copied the hashes of the users from the victim machine to the attack machine for cracking




I view the content copied from the ubuntu (victim) machine and 3 hashes was discovered for three users: alice, bob and charlie




I attempted a dictionary attack on the extracted file using hashcat

kali linux (after upgrade) [Running] - Oracle VM VirtualBox

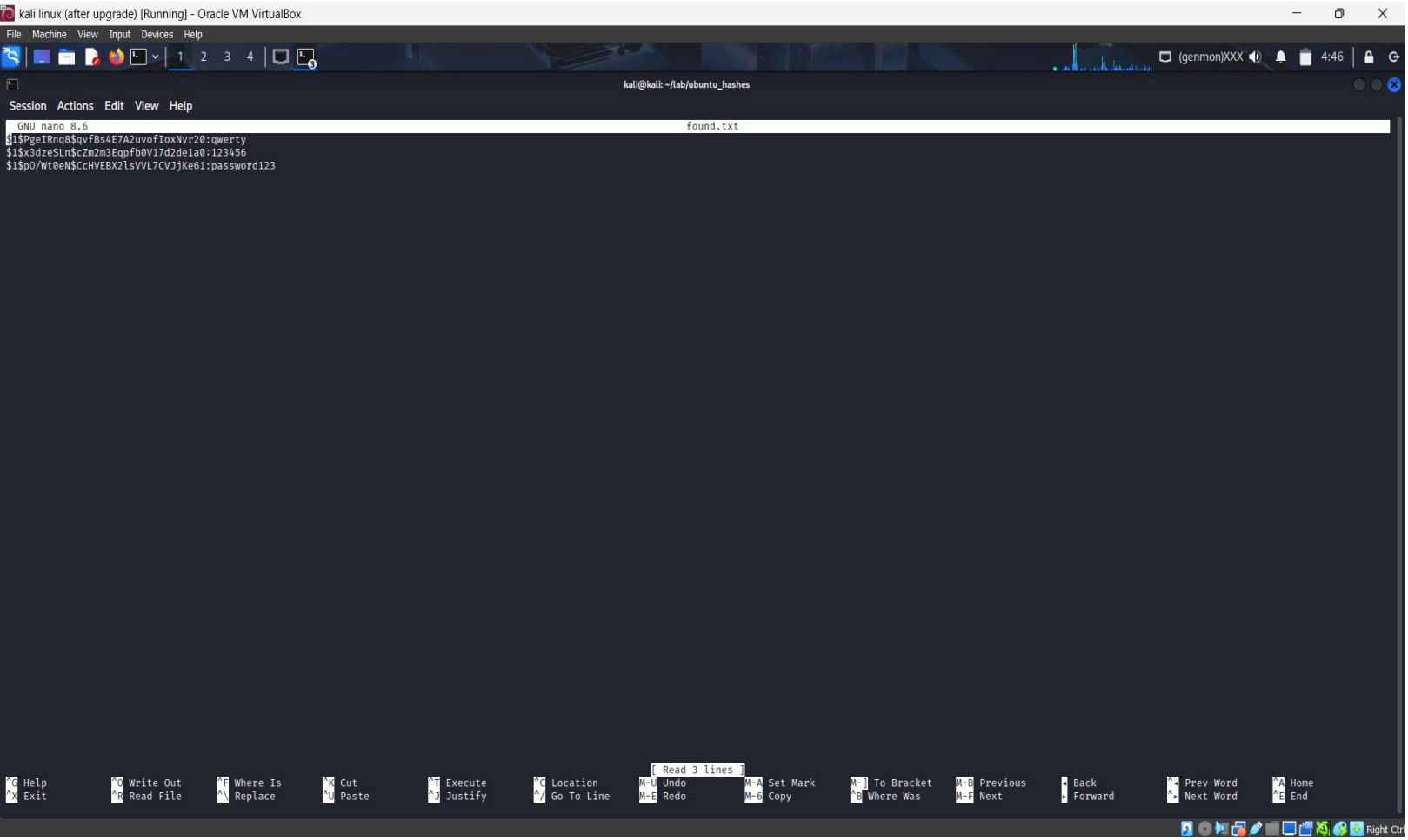
File Machine View Input Devices Help

kali@kali: ~/lab/ubuntu_hashes

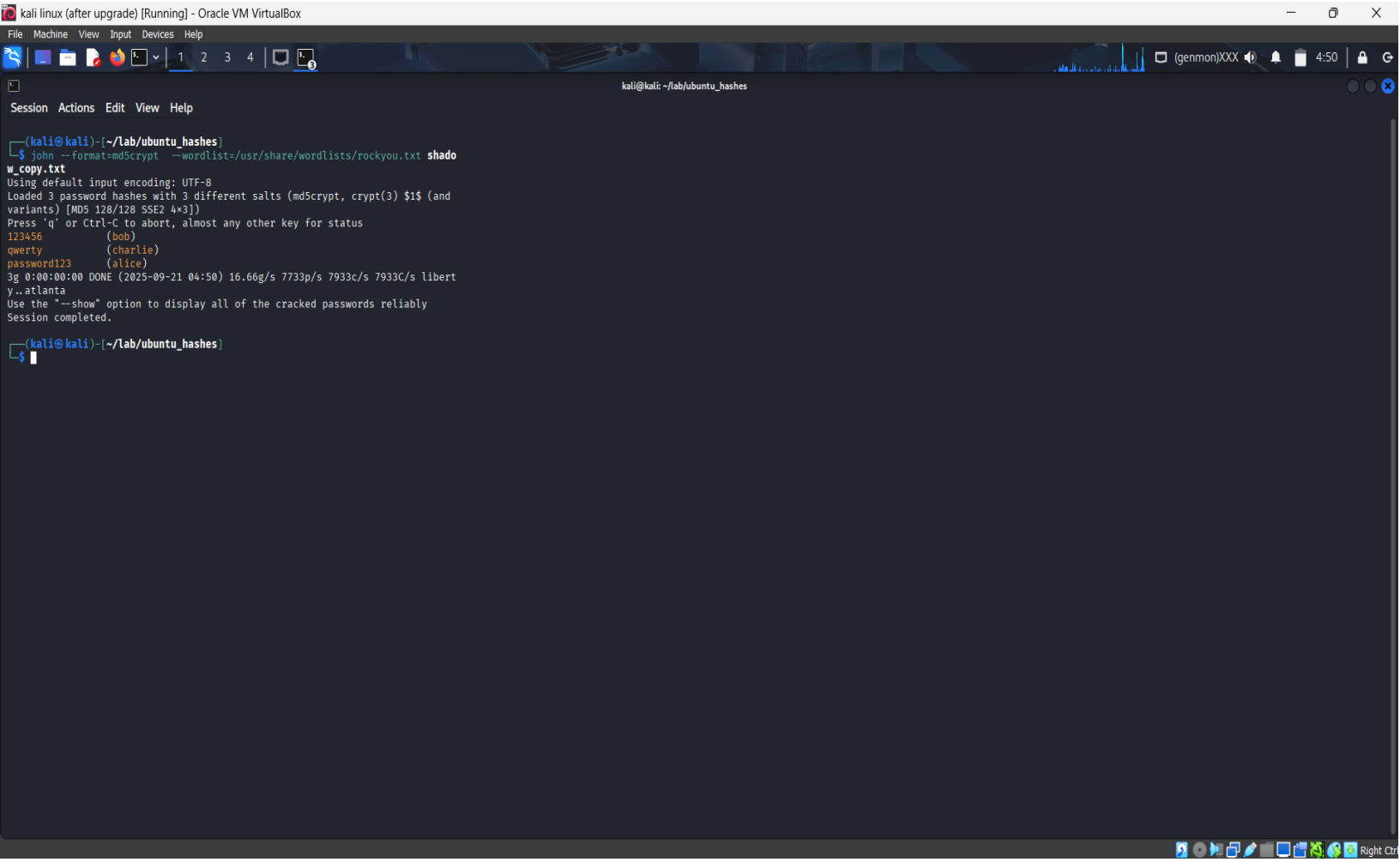
Session Actions Edit View Help

```
┌──(kali㉿kali)-[~/lab/ubuntu_hashes]
└─$ hashcat -m 500 -a 0 --username shadow_copy.txt /usr/share/wordlists/rockyou.txt -o found.txt --status --status-timer=10
hashcat (v6.2.6) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian  Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]
====================================================================================================================================
* Device #1: cpu-penryn-Intel(R) Core(TM) i7-8550U CPU @ 1.80GHz, 1085/2234 MB (512 MB allocatable), 1MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

INFO: All hashes found as potfile and/or empty entries! Use --show to display them.

Started: Sun Sep 21 04:45:21 2025
Stopped: Sun Sep 21 04:45:22 2025

┌──(kali㉿kali)-[~/lab/ubuntu_hashes]
└─$ ls
found.txt  shadow_copy.txt

┌──(kali㉿kali)-[~/lab/ubuntu_hashes]
└─$
```

After a successful attack the result was saved in the file found.txt




After which I also attempted to use john the ripper to crack the hash file




# Challenges

1. Mis-identification of correct hash mode for hashcat: inspected hash prefix ($1$, $5$ etc)
2. Permission copy error for /etc/shadow: in lab elevate with sudo/root

# Mitigation Recommnedation

1. Enforce password policy (minimum of 12 characters, use of combinations of letters, numbers, special character, upper case and lower case)
2. Block commonly used password
3. Enable multi-factor authentication for interactive logins
4. Close unused ports